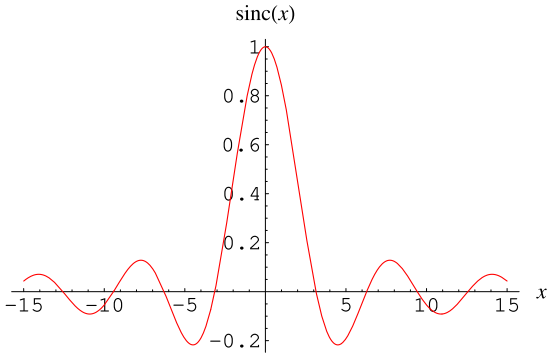
Sinc Function

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The sinc function $\operatorname{sinc}(x)$, also called the "sampling function," is a function that arises frequently in signal processing and the theory of Fourier transforms. The full name of the function is "sine cardinal," but it is commonly referred to by its abbreviation, "sinc." There are two definitions in common use. The one adopted in this work defines

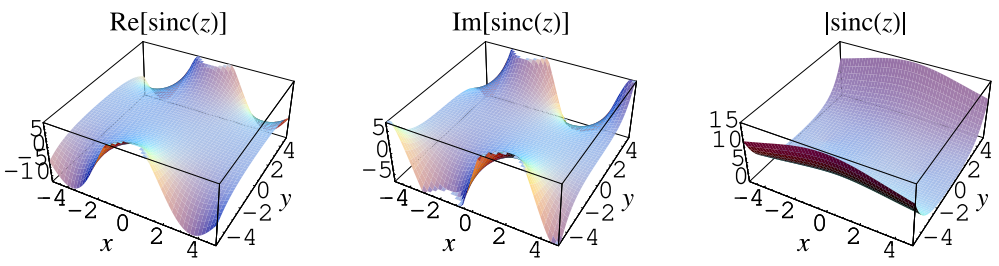
$$\operatorname{sinc}(x) \equiv \begin{cases} 1 & \text{for } x = 0 \\ \dfrac{\sin x}{x} & \text{otherwise,} \end{cases} \tag{1}$$

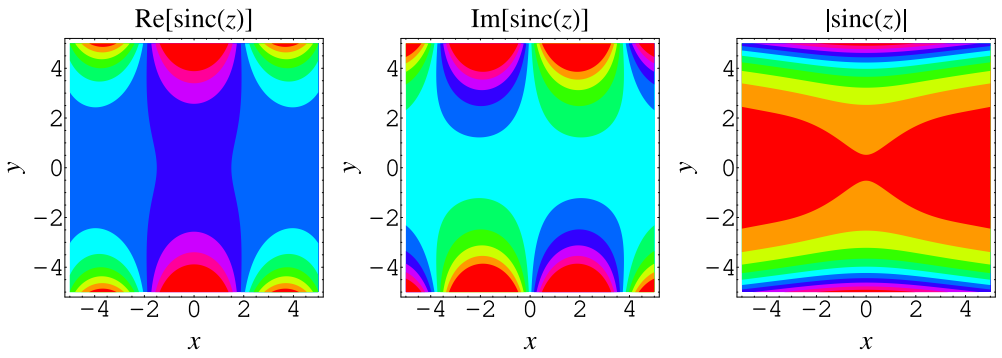
where $\sin x$ is the sine function, plotted above.

This has the normalization

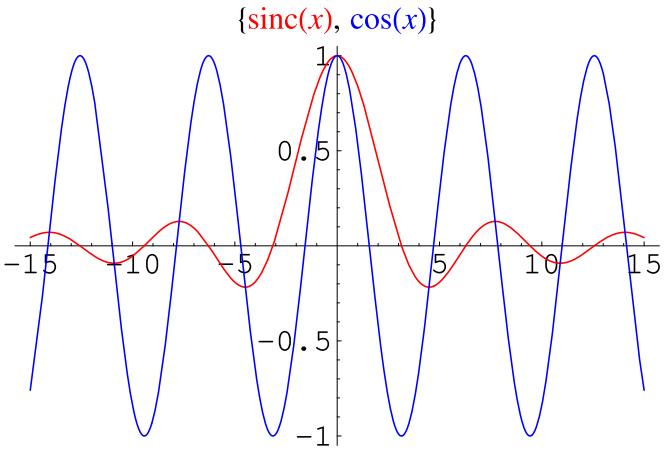
$$\int_{-\infty}^{\infty} \operatorname{sinc}(x)\, dx = \pi. \tag{2}$$

This function is implemented in the Wolfram Language as Sinc[x].





When extended into the complex plane, $\operatorname{sinc}(z)$ is illustrated above.



An interesting property of $\operatorname{sinc}(x)$ is that the set of local extrema of $\operatorname{sinc}(x)$ corresponds to its intersections with the cosine function $\cos(x)$, as illustrated above.

The derivative is given by

$$\frac{d\operatorname{sinc}(z)}{dz} = \frac{\cos z}{z} - \frac{\sin z}{z^2} \tag{3}$$

and the indefinite integral by

$$\int \operatorname{sinc}(z)\,dz = \operatorname{Si}(z), \tag{4}$$

where $\operatorname{Si}(z)$ is the sine integral.

Woodward (1953), McNamee *et al.* (1971), and Bracewell (1999, p. 62) adopt the alternative definition

$$\operatorname{sinc}_\pi(x) \equiv \operatorname{sinc}(\pi x) \tag{5}$$

$$= \begin{cases} 1 & \text{for } x = 0 \\ \dfrac{\sin(\pi x)}{\pi x} & \text{otherwise.} \end{cases} \tag{6}$$

The latter definition is sometimes more convenient as a result of its simple normalization,

$$\int_{-\infty}^{\infty} \operatorname{sinc}_\pi(x)\,dx = 1. \tag{7}$$

That variant also satisfies the sum

$$\sum_{k=-\infty}^{\infty} \operatorname{sinc}(\pi k) = 1. \tag{8}$$

In addition, the binomial coefficient satisfies

$$\binom{0}{x} = \operatorname{sinc}_{pi}(x), \tag{9}$$

which is essentially a restatement of the reflection relation

$$\Gamma(1+x)\,\Gamma(1-x) = \frac{\pi x}{\sin(\pi x)} = \frac{1}{\operatorname{sinc}_{pi}(x)} \tag{10}$$

of the gamma function (M. Somos, pers. comm., Oct. 26, 2006.)

The sinc function is closely related to the spherical Bessel function of the first kind $j_n(x)$ and, in particular,

$$\operatorname{sinc}(x) = j_0(x), \tag{11}$$

and is given in terms of the Meijer G-function as

$$\operatorname{sinc}(z) = \frac{\sqrt{\pi}}{2}\, G_{02}^{10}\left(\tfrac{1}{2}\, z,\ \tfrac{1}{2}\ \middle|\ \begin{matrix} - \\ 0,\ \tfrac{1}{2} \end{matrix}\right). \tag{12}$$

Let $\Pi(x)$ be the rectangle function, then the Fourier transform of $\Pi(x)$ is the sinc function

$$\mathcal{F}_x[\Pi(x)](k) = \operatorname{sinc}(\pi k). \tag{13}$$

The sinc function therefore frequently arises in physical applications such as Fourier transform spectroscopy as the so-called instrument function, which gives the instrumental response to a delta function input. Removing the instrument functions from the final spectrum requires use of some sort of deconvolution algorithm.

The sinc function can be written as a complex integral by noting that, for $x \neq 0$,

$$\operatorname{sinc}(n\,x) \equiv \frac{\sin(n\,x)}{n\,x} \tag{14}$$

$$= \frac{1}{n\,x}\,\frac{e^{inx} - e^{-inx}}{2\,i} \tag{15}$$

$$= \frac{1}{2\,i\,n\,x}\left[e^{itx}\right]_{-n}^{n} \tag{16}$$

$$= \frac{1}{2\,n}\int_{-n}^{n} e^{ixt}\,dt, \tag{17}$$

and that $\operatorname{sinc}(n\,x)$ and the integral both equal 1 for $x = 0$.

The sinc function can also be written as the infinite product

$$\operatorname{sinc}(x) = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right), \tag{18}$$

a result discovered in 1593 by Francois Viète (Kac 1959, Morrison 1995) and sometimes known as Euler's formula (Prudnikov *et al.* 1986, p. 757; Gearhart and Shulz 1990). It is also given by

$$\operatorname{sinc}(x) = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2\,\pi^2}\right) \tag{19}$$

(Gearhart and Shulz 1990) and

$$\operatorname{sinc}(x) = \prod_{k=1}^{\infty}\left[1 - \frac{4}{3}\sin^2\left(\frac{x}{3^k}\right)\right] \tag{20}$$

(Prudnikov *et al.* 1986, p. 757).

Another product is given by

$$\prod_{k=1}^{\infty}\operatorname{sinc}\left(\frac{2\,\pi}{2\,k+1}\right) = \frac{\pi}{2\,K} \tag{21}$$

$$= 0.1805504\ldots \tag{22}$$

(OEIS A118253; Prudnikov *et al.* 1986, p. 757), where $K = 8.700\ldots$ is the constant from polygon circumscribing.

Sums of powers of $\operatorname{sinc}(k)$ over the positive integers include

$$\sum_{k=1}^{\infty}\operatorname{sinc}(k) = -\tfrac{1}{2} + \tfrac{1}{2}\,\pi \tag{23}$$

$$\sum_{k=1}^{\infty}\operatorname{sinc}^2(k) = -\tfrac{1}{2} + \tfrac{1}{2}\,\pi \tag{24}$$

$$\sum_{k=1}^{\infty}\operatorname{sinc}^3(k) = -\tfrac{1}{2} + \tfrac{3}{8}\,\pi \tag{25}$$

$$\sum_{k=1}^{\infty}\operatorname{sinc}^4(k) = -\tfrac{1}{2} + \tfrac{1}{3}\,\pi \tag{26}$$

$$\sum_{k=1}^{\infty}\operatorname{sinc}^5(k) = -\tfrac{1}{2} + \tfrac{115}{384}\,\pi \tag{27}$$

$$\sum_{k=1}^{\infty}\operatorname{sinc}^6(k) = -\tfrac{1}{2} + \tfrac{11}{40}\,\pi. \tag{28}$$

The remarkable fact that the sums of $\operatorname{sinc}(k)$ and $\operatorname{sinc}(k)^2$ are equal appears to have first been published in Baillie (1978). Amazingly, the pattern of these sums being equal to $-1/2$ plus a rational multiple of π breaks down for the power $n = 7$, where the sum equals

$$\sum_{k=1}^{\infty}\operatorname{sinc}^7(k) = -\tfrac{1}{2} + P(x), \tag{29}$$

where

$$P(x) = \tfrac{1}{46\,080}\,\pi\left(129\,423 - 201\,684\,\pi + 144\,060\,\pi^2 - 54\,880\,\pi^3 + 11\,760\,\pi^4 - 1344\,\pi^5 + 64\,\pi^6\right). \tag{30}$$

The sinc function satisfies the identity

$$\int_{-\infty}^{\infty} \operatorname{sinc}\left[\pi\left(x-y\right)\right] \operatorname{sinc}\left(\pi\, y\right) d\, y = \operatorname{sinc}\left(\pi\, x\right). \tag{31}$$

Definite integrals involving the sinc function include

$$\int_0^{\infty} \operatorname{sinc}\left(x\right) d\, x = \tfrac{1}{2}\, \pi \tag{32}$$

$$\int_0^{\infty} \operatorname{sinc}^2\left(x\right) d\, x = \tfrac{1}{2}\, \pi \tag{33}$$

$$\int_0^{\infty} \operatorname{sinc}^3\left(x\right) d\, x = \tfrac{3}{8}\, \pi \tag{34}$$

$$\int_0^{\infty} \operatorname{sinc}^4\left(x\right) d\, x = \tfrac{1}{3}\, \pi \tag{35}$$

$$\int_0^{\infty} \operatorname{sinc}^5\left(x\right) d\, x = \tfrac{115}{384}\, \pi. \tag{36}$$

After dividing out the constant factor of π, the values are again 1/2, 1/2, 3/8, 1/3, 115/384, 11/40, 5887/23040, 151/630, 259723/1146880, ... (OEIS A049330 and A049331; Grimsey 1945, Medhurst and Roberts 1965). These are all special cases of the amazing general result

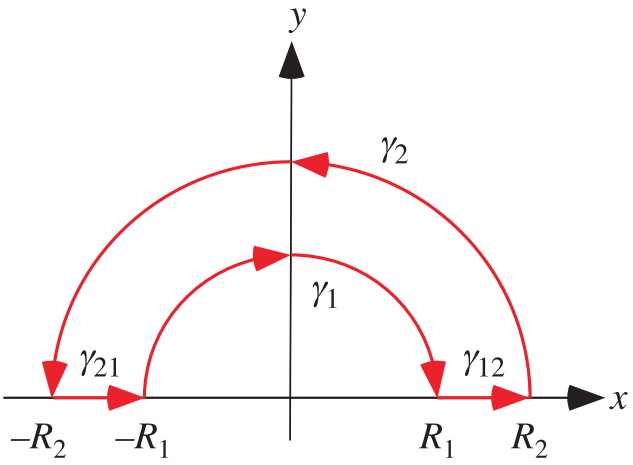
$$\int_0^{\infty} \frac{\sin^a x}{x^b}\, d\, x = \frac{\pi^{1-c}\, (-1)^{\lfloor (a-b)/2 \rfloor}}{2^{a-c}\, (b-1)\,!} \sum_{k=0}^{\lfloor a/2 \rfloor - c} (-1)^k \binom{a}{k} (a-2\, k)^{b-1}\, [\ln\left(a-2\, k\right)]^c, \tag{37}$$

where a and b are positive integers such that $a \geq b > c$, $c \equiv a - b \pmod 2$, $\lfloor x \rfloor$ is the floor function, and 0^0 is taken to be equal to 1 (Kogan; cf. Espinosa and Moll 2000). This spectacular formula simplifies in the special case when n is a positive even integer to

$$\int_0^{\infty} \frac{\sin^{2\, n} x}{x^{2\, n}}\, d\, x = \frac{\pi}{2\, (2\, n-1)\,!} \left\langle {2\, n-1 \atop n-1} \right\rangle, \tag{38}$$

where $\left\langle {n \atop k} \right\rangle$ is an Eulerian number (Kogan; cf. Espinosa and Moll 2000). The solution of the integral can also be written in terms of the recurrence relation for the coefficients

$$c\left(a,\, b\right) = \begin{cases} \dfrac{\pi}{2^{a+1-b}} \begin{pmatrix} a-1 \\ \frac{1}{2}\, (a-1) \end{pmatrix} & \text{for } b = 1 \text{ or } b = 2 \\ \dfrac{a\, [(a-1)\, c\left(a-2,\, b-2\right) - a \cdot c\left(a,\, b-2\right)]}{(b-1)\, (b-2)} & \text{otherwise.} \end{cases} \tag{39}$$



The half-infinite integral of $\operatorname{sinc}(x)$ can be derived using contour integration. In the above figure, consider the path $\gamma \equiv \gamma_1 + \gamma_{12} + \gamma_2 + \gamma_{21}$. Now write $z = R\,e^{i\,\theta}$. On an arc, $d\,z = i\,R\,e^{i\,\theta}\,d\,\theta$ and on the x-axis, $d\,z = e^{i\,\theta}\,d\,R$. Write

$$\int_{-\infty}^{\infty} \operatorname{sinc}(x)\,d\,x = \Im \int_{\gamma} \frac{e^{i\,z}}{z}\,d\,z, \tag{40}$$

where $\Im$ denotes the imaginary part. Now define

$$I \equiv \int_{-\infty}^{\infty} \frac{e^{i\,z}}{z}\,d\,z \tag{41}$$

$$= \lim_{R_1 \to 0} \int_{\pi}^{0} \frac{\exp\left(i\,R_1\,e^{i\theta}\right)}{R_1\,e^{i\theta}}\,i\,R_1\,e^{i\theta}\,d\,\theta + \lim_{R_1 \to 0} \lim_{R_2 \to \infty} \int_{R_1}^{R_2} \frac{e^{i\,R}}{R}\,d$$

$$R + \lim_{R_2 \to \infty} \int_{0}^{\pi} \frac{\exp(i\,z)}{z}\,d\,z + \lim_{\substack{R_1 \to 0 \\ R_2 \to \infty}} \int_{R_2}^{R_1} \frac{e^{-i\,R}}{-R}\,(-d\,R), \tag{42}$$

where the second and fourth terms use the identities $e^{i\,0} = 1$ and $e^{i\,\pi} = -1$. Simplifying,

$$I = \lim_{R_1 \to 0} \int_{\pi}^{0} \exp\left(i\,R_1\,e^{i\theta}\right) i\,d\,\theta + \int_{0^+}^{\infty} \frac{e^{i\,R}}{R}\,d$$

$$R + \lim_{R_2 \to \infty} \int_{0}^{\pi} \frac{\exp(i\,z)}{z}\,d\,z + \int_{\infty}^{0^+} \frac{e^{-i\,R}}{-R}\,(-d\,R) \tag{43}$$

$$= -\int_{0}^{\pi} i\,d\,\theta + \int_{0^+}^{\infty} \frac{e^{i\,R}}{R}\,d\,R + 0 + \int_{-\infty}^{0^-} \frac{e^{i\,R}}{R}\,d\,R, \tag{44}$$

where the third term vanishes by Jordan's lemma. Performing the integration of the first term and combining the others yield

$$I = -i\,\pi + \int_{-\infty}^{\infty} \frac{e^{i\,z}}{z}\,d\,z = 0. \tag{45}$$

Rearranging gives

$$\int_{-\infty}^{\infty} \frac{e^{i\,z}}{z}\,d\,z = i\,\pi, \tag{46}$$

so

$$\int_{-\infty}^{\infty} \frac{\sin z}{z}\, dz = \pi. \tag{47}$$

The same result is arrived at using the method of complex residues by noting

$$I = 0 + \tfrac{1}{2}\, 2\pi i \operatorname*{Res}_{z=0} f(z) \tag{48}$$

$$= i\pi (z-0)\, \frac{e^{iz}}{z}\bigg|_{z=0} \tag{49}$$

$$= i\pi \left[e^{iz}\right]_{z=0} \tag{50}$$

$$= i\pi, \tag{51}$$

so

$$\mathfrak{I}(I) = \pi. \tag{52}$$

Since the integrand is symmetric, we therefore have

$$\int_0^{\infty} \frac{\sin x}{x}\, dx = \tfrac{1}{2}\pi, \tag{53}$$

giving the sine integral evaluated at 0 as

$$\operatorname{si}(0) = -\int_0^{\infty} \frac{\sin x}{x}\, dx = -\tfrac{1}{2}\pi. \tag{54}$$

SEE ALSO

Borwein Integrals, Fourier Transform, Fourier Transform--Rectangle Function, Instrument Function, Jinc Function, Kilroy Curve, Sine, Sine Integral, Sinhc Function, Tanc Function

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